CPSY 1291 · Recitation handout 14

Project clinic II

Figures that are instruments, a report that is honest, and a talk that lands

Fall 2026 · Week 14 · last class meeting Thu 12/10; poster session (report due) Mon 12/21, 9:00 AM

What this session is for

By now you have results, or you have discovered that you will not get the ones you wanted. Both are fine, and the syllabus is explicit that a negative result clearly analyzed is a perfectly good project. What is graded is the quality of the question, the soundness of the method, and how honestly you interpret what you found.

This handout is about turning what you have into three things: figures that carry an argument, a report that survives a skeptical reader, and a talk that lands in the time you are given.

> **The one idea.** **The claim comes first, and everything else is evidence for it.** Write your one-sentence result before you make a single figure. Then every figure either supports that sentence or comes out. Projects that assemble figures first and look for a story afterwards produce talks that are a tour of what was done rather than an argument about what is true.

1 Figures as instruments

1.1 One figure, one claim

Each figure should be describable in one sentence that is a claim, not a description. "Accuracy by condition and model" is a description. "The texture model loses more accuracy under shape distortion than the shape model" is a claim, and it tells you what to plot and what to leave out.

If a figure needs two sentences, it is two figures. If it needs none, it is not a figure.

1.2 The checklist

Every figure	Why
axis labels with **units**	an unlabeled axis is uninterpretable
error bars, and a caption saying which	SD and SE answer different questions
a **chance line** where one exists	a reader cannot infer it
n stated — of *what*	runs? stimuli? subjects?
readable at the size it will be seen	shrink it to slide size and check
distinguishable without color	about 8% of men are color-blind

The error-bar rule, once more. Label them. *Standard deviation* says how much a single run varies; *standard error* says how well you know the mean. An unlabeled error bar cannot be read, and a figure whose error bars are unlabeled is asking the reader to trust you rather than to check you.

1.3 Three plots that are almost always worth making

The distribution, not just the mean. A bar chart of two means hides whether the distributions overlap completely. Plot the individual points over the bars — with ten runs there is no reason not to.

The thing that would falsify you. If your claim is that model A beats model B, plot the per-item differences, not just the two averages. If A wins on 51% of items by a hair, that is a different claim from A winning on 90%.

The failure cases. Show the items your model got wrong. It is the most informative figure in most projects and it is almost always missing, because nobody wants to put their failures on a slide. It is also what makes a talk credible.

1.4 Two things not to do

Do not truncate a y-axis to make a difference look big. If the difference needs a truncated axis, the difference is small, and saying so is better than being caught.

Do not use a rainbow colormap for continuous data. It creates boundaries that are not in the data. `viridis` is the default for a reason.

2 The report

2.1 A structure that always works

Section	The one job it does
Question	what you asked, and why it is worth asking
Method	enough that someone else could redo it
Results	what happened — figures, numbers, no interpretation
Interpretation	what it means, and what it does not
Limitations	what would change your mind

The split between Results and Interpretation is the one that matters. Keeping them separate is what lets a reader disagree with your interpretation while accepting your results, which is how science is supposed to work — and it is what makes a report readable by someone who does not already agree with you.

2.2 Limitations is a strength section

Students write limitations defensively, as damage control. Written well it is the section that demonstrates you understand your own work.

Say what would change your mind. "We used one architecture, so we cannot separate an effect of depth from an effect of this particular network" is a strong sentence: it names the confound, bounds the claim, and tells the reader exactly which experiment would settle it. "More data would help" is not a limitation; it is a truism.

2.3 The AI disclosure

The syllabus permits AI tools for the project and requires a short disclosure saying which tools you used and how they contributed. Write it plainly and specifically — "we used an LLM to draft the plotting code and to debug the data loader" — rather than vaguely. It costs two lines, it is a course requirement, and vagueness reads worse than the use itself.

3 The talk

3.1 The arc

Slide(s)	Job
1	the question, in one sentence, and why anyone should care
2	what you did, in one picture — not in bullets
3–5	the result: one figure per slide, one claim per figure
6	the honest caveat: what would change your mind
7	what you would do next

> **Slide 2 is the one people get wrong.** Your method is a pipeline, and a pipeline is a picture — boxes and arrows, from data to number. Three bullet points describing the same pipeline take longer to say and are harder to follow. Draw it.

3.2 Rehearsal, which is not optional

1. **Out loud, with a timer, once.** Silent reading takes roughly half the time speaking does, so a deck that feels right unspoken is about twice too long.
2. **Decide who says what.** In a group talk, silent handovers are the commonest thing to go wrong. Write the handover into the notes.
3. **Cut to fit.** If you are over, cut a *result*, not the caveat and not the question. A talk that covers less and lands is better than one that covers everything at speed.

3.3 Questions you will be asked

- *What is chance here?* — have the number, not a gesture.
- *How many seeds / subjects / stimuli?* — have the n, and say what it is the n *of*.
- *What does a trivial baseline get?* — if you do not have one, say so plainly; do not improvise.

- *Would this hold for a different dataset or architecture?* — the honest answer is usually "we do not know, and here is why we think it might not".
- *Why this metric?* — have a reason, even a modest one.

> **"We don't know" is a good answer.** Said once, with a reason, it reads as command of the material. What reads badly is a confident answer to a question you did not investigate — and the room can tell the difference.

4 If the result did not come out

This happens, and it is survivable. The syllabus grades the question, the method and the honesty of the interpretation — not whether the model worked.

1. **Say what you expected and what you got.** Both, plainly, early.
2. **Show that the pipeline works.** A positive control — a case where your method *does* detect a known effect — turns "our analysis found nothing" into "there is nothing there to find, and here is the evidence that we could have found it". Those are very different claims and the second is a result.
3. **Bound it.** How large an effect could you have detected? A null with a stated sensitivity is informative; a null without one is silence.
4. **Say what you would do differently.** This is the part that shows you learned something, which is what the exercise was for.

5 The last checklist

Run this the night before.

1. Can you state the result in one sentence, without hedging?
2. Does every figure have labeled axes, labeled error bars, and a stated n?
3. Is chance marked on every figure that has one?
4. Does the talk have a positive control or a trivial baseline somewhere?
5. Have you said out loud what would change your mind?
6. Did you rehearse with a timer, out loud, at least once?
7. Does the report separate what happened from what you think it means?
8. Is the AI-use disclosure written?
9. Can someone else run your pipeline from your loader?
10. Have you cut the slide you are keeping only because the work was hard?

Item 10 catches something in almost every project.